

P1: Stationary Points, Goldstone Alignment, Complete Hessian, and Spectrum in P54PQ-v2

Route-F four-dimensional mainline

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Abstract

We construct P1 directly from the corrected P54PQ-v2 tensor action. The differentiable field space has $54 + 2(126) + 2(10) + 2 = 328$ canonically normalized real coordinates. We derive the radial stationary equations in the action card's own normalization, enumerate competing radial branches, construct the complete 328×328 scalar Hessian by automatic differentiation, and independently construct the 45-generator gauge orbit. Gauge invariance requires 33 gauge zero modes; the faithful PQ symmetry adds one physical axion direction after projecting away the gauge orbit. The calculation is fail-closed: a stationary saddle is reported as a saddle, and the electroweak doublet tuning is not silently identified with vacuum stability.

1 Why P1 is restarted on version 2

The independent invariant audit found

$$V_7 = \chi_7 \Phi_{ij} \phi_i \phi_j S^* + \chi_7^* \Phi_{ij} \phi_i^* \phi_j^* S. \quad (1)$$

It contributes at second order in the 10_H fluctuation whenever $\langle \Phi \rangle$ and $\langle S \rangle$ are nonzero. Therefore it changes the 10_H diagonal and holomorphic mass blocks even though it does not alter the high-scale radial stationarity equations at $\langle \phi \rangle = 0$. Computing a “complete” Hessian from version 1 and appending (1) after diagonalization would be mathematically invalid.

The earlier literature potential also contains printed conjugation and index errors, and its displayed radial coefficients are not all reproduced by its displayed factorials under a single tensor normalization. P1 consequently uses one source of truth: the explicit tensor action in the corrected action card. Literature mass tables are comparisons, not definitions.

2 Canonical field coordinates

Let E_{ij}^a , $a = 1, \dots, 54$, be an orthonormal basis of real symmetric traceless matrices,

$$E_{ij}^a E_{ij}^b = \delta^{ab}, \quad \Phi_{ij} = x_a E_{ij}^a. \quad (2)$$

Let $U_{A\alpha}$ be an orthonormal basis of one Hodge eigenspace in the 252 independent five-form components, $\alpha = 1, \dots, 126$. The complex fields are written

$$\begin{aligned} z_\alpha &= \frac{x_{54+\alpha} + ix_{180+\alpha}}{\sqrt{2}}, & \Sigma_A &= U_{A\alpha} z_\alpha, \\ h_i &= \frac{x_{306+i} + ix_{316+i}}{\sqrt{2}}, & S &= \frac{x_{326} + ix_{327}}{\sqrt{2}}. \end{aligned} \quad (3)$$

In these variables the kinetic term is $\frac{1}{2} \partial x_I \partial x_I$. The full scalar coordinate count is

$$\boxed{N_{\mathbb{R}} = 54 + 252 + 20 + 2 = 328.} \quad (4)$$

The high-scale background is

$$\begin{aligned}\Phi_0 &= \omega X, & X &= \text{diag}(-2/5, -2/5, -2/5, -2/5, -2/5, -2/5, 3/5, 3/5, 3/5, 3/5), \\ \Sigma_0 &= \sigma \Omega, & qquad \|\Omega\|_{\Lambda^5}^2 &= 1, & qquad S_0 &= \frac{v_s}{\sqrt{2}}, & h_0 &= 0.\end{aligned}\tag{5}$$

The ordered component of the holomorphic form has magnitude $|\Omega_{246810}| = 1/(4\sqrt{2})$, matching the action card.

3 Stationary equations in the card normalization

Direct substitution of (5) into the tensor action gives

$$\begin{aligned}V_{\text{rad}} &= -\frac{6}{5}\mu^2\omega^2 + \frac{4}{25}c\omega^3 + A_4\omega^4 - \frac{1}{2}\nu^2\sigma^2 + \frac{1}{4}\lambda_0\sigma^4 + \frac{6}{5}d\omega^2\sigma^2 \\ &\quad - \frac{1}{2}\mu_s^2v_s^2 + \frac{1}{4}\chi_1v_s^4 + 60\chi_2\sigma^2v_s^2 + \frac{6}{5}\chi_3\omega^2v_s^2, \quad A_4 = \frac{36}{25}a + \frac{42}{125}b, & qquad d &= \alpha - \beta.\end{aligned}\tag{6}$$

The $\lambda_{2,4,4'}$ and χ_4 contractions vanish on the decomposable holomorphic five-form, but their second variations do not vanish and they remain in the full Hessian. Equation (6) is not imported from a radial formula; it is an executable contraction fingerprint of the card.

For the fully nonzero branch, stationarity is equivalent to

$$\mu^2 = \frac{1}{5}c\omega + \frac{5}{3}A_4\omega^2 + d\sigma^2 + \chi_3v_s^2,\tag{7}$$

$$\nu^2 = \lambda_0\sigma^2 + \frac{12}{5}d\omega^2 + 120\chi_2v_s^2,\tag{8}$$

$$\mu_s^2 = \chi_1v_s^2 + 120\chi_2\sigma^2 + \frac{12}{5}\chi_3\omega^2.\tag{9}$$

The factors of 60 and 120 are not physical enhancements; they are the declared normalization of the coefficient χ_2 in the tensor card. A rescaled coefficient $\hat{\chi}_2 = 120\chi_2$ would remove them, but mixing the two parameter conventions is forbidden.

The radial curvature before conversion to the canonical metric is

$$K_{\text{rad}} = \begin{pmatrix} \frac{12}{25}c\omega + 8A_4\omega^2 & \frac{24}{5}d\omega\sigma & \frac{24}{5}\chi_3\omega v_s \\ \frac{24}{5}d\omega\sigma & 2\lambda_0\sigma^2 & 240\chi_2\sigma v_s \\ \frac{24}{5}\chi_3\omega v_s & 240\chi_2\sigma v_s & 2\chi_1v_s^2 \end{pmatrix}.\tag{10}$$

Its generalized eigenvalues use $G_{\text{rad}} = \text{diag}(12/5, 2, 1)$, since the coordinates (ω, σ, v_s) are not all canonical. Positivity of (10) is necessary but not sufficient for a vacuum.

Every subset of $\{\omega, \sigma, v_s\}$ can vanish, so the radial problem has eight algebraic branch types before sign and multiplicity are resolved. The machine audit starts from a grid of positive, negative, and zero initial conditions, deduplicates all roots, evaluates V_{rad} , and records the three radial curvatures. This prevents the desired nonzero branch from being declared the vacuum merely because its mass squares were solved from (7)–(9).

4 Complete Hessian

The scalar Hessian is defined without a representation-level truncation:

$$\boxed{\mathcal{H}_{IJ} = \left. \frac{\partial^2 V}{\partial x_I \partial x_J} \right|_{x=x_0}}, \quad I, J = 1, \dots, 328.\tag{11}$$

Automatic differentiation is applied to every indexed contraction in $V_{\Phi\Sigma} + V_{\Phi\Sigma\phi} + V_S$, including all Hermitian partners. The block layout is

$$\mathcal{H} = \begin{pmatrix} H_{54,54} & H_{54,126} & H_{54,10} & H_{54,S} \\ H_{126,54} & H_{126,126} & H_{126,10} & H_{126,S} \\ H_{10,54} & H_{10,126} & H_{10,10} & H_{10,S} \\ H_{S,54} & H_{S,126} & H_{S,10} & H_{S,S} \end{pmatrix}_{(54+252+20+2) \times (54+252+20+2)}. \quad (12)$$

At $h_0 = 0$, symmetry makes some displayed blocks vanish, but they are computed rather than omitted. In particular, η_1 can mix the 126 and 10 sectors at quadratic order, χ_4 mixes singlet sectors, and (1) changes the 10 block. The verifier repeats the Hessian with $\chi_7 = 0$ and requires a nonzero Frobenius difference.

5 Goldstone alignment theorem and its numerical form

For a generator T_A , gauge invariance says

$$0 = \frac{d}{dt} V(e^{tT_A} x) \Big|_{t=0} = \partial_I V(x) (T_A x)^I. \quad (13)$$

Differentiating with respect to x^J and imposing stationarity gives

$$\mathcal{H}_{JI} (T_A x_0)^I = 0. \quad (14)$$

Thus the theorem compares actual vectors, not just the number of small eigenvalues. The 45-column orbit matrix is

$$\mathcal{M}_G = (T_1 x_0 \ \cdots \ T_{45} x_0). \quad (15)$$

The independent tensor construction gives

$$\text{rank } \mathcal{M}_G(\Phi_0) = 24, \text{quadrank } \mathcal{M}_G(\Phi_0, \Sigma_0) = 33, \quad (16)$$

so the unbroken algebra has dimension $45 - 33 = 12$, as required for $SU(3)_C \times SU(2)_L \times U(1)_Y$.

Let qx_0 be the infinitesimal PQ direction. Its component parallel to the gauge orbit is unphysical, so the physical axion direction is

$$a_0 = [1 - \mathcal{M}_G \mathcal{M}_G^+] qx_0, \quad (17)$$

where $+$ denotes the Moore–Penrose inverse. P1 requires

$$\text{rank}(\mathcal{M}_G, a_0) = 34, \text{quad} \frac{\|\mathcal{H} \mathcal{M}_G\|}{\|\mathcal{H}\| \|\mathcal{M}_G\|} \ll 1, \text{quad} \frac{\|\mathcal{H} a_0\|}{\|\mathcal{H}\| \|a_0\|} \ll 1. \quad (18)$$

The QCD anomaly does not enter this tree-level Hessian; it lifts the axion only in the low-energy nonperturbative effective action.

6 Scalar and vector spectra

The scalar spectrum is the complete eigenvalue multiset

$$\text{Spec}_{\text{scalar}} = \text{eigvalsh}(\mathcal{H}). \quad (19)$$

Degeneracies are grouped only after the Goldstone residual fixes an absolute tolerance. A negative physical eigenvalue classifies the background as a stationary saddle even when the radial 3×3 block is positive.

The vector spectrum follows from the same orbit matrix and not from the potential:

$$\frac{M_V^2}{g_{10}^2} = \mathcal{M}_G^T \mathcal{M}_G. \quad (20)$$

Its nullity must be 12 and its positive rank 33. Equation (20) also fixes the generator normalization and prevents an arbitrary rescaling of the scalar Goldstone vectors.

This P1 benchmark is dimensionless and exercises every Hessian block; it is not a flavor, unification, or proton-decay fit. It has no negative physical scalar mode and exactly one tuned complex electroweak doublet. The doublet tuning is a rank condition on the full mixed doublet block, not an independent free mass.

7 Executed benchmark and complete spectrum

We measure all dimensionful quantities in units of $\omega = 1$ and choose

$$\begin{aligned} (\sigma, v_s) &= (0.35, 0.25), & (a, b, c) &= (0.50, 0.40, -0.10), \\ (\lambda_0, \lambda_2, \lambda_4, \lambda'_4) &= (0.90, 0.80, 0.60, 0.40), & (\alpha, \beta) &= (0.25, 0.10), \\ (\chi_1, \chi_2, \chi_3, \chi_4, \chi_5, \chi_6, \chi_7) &= (0.80, 0.001, 0.05, 0.03, 0.20, 0.025, 0.03), \\ (\eta_0, \eta_1, \eta_2, \eta_3) &= (0.20, 0.025, 0.10, 0.018), & (\gamma_1, \gamma_2) &= (0.12, -0.07), \\ (\xi_1, \xi_2, \xi_3) &= (0.35, 0.08, -0.20). \end{aligned}$$

Equations (7)–(9) then give

$$(\mu^2, \nu^2, \mu_s^2) = (1.4255, 0.47775, 0.1847). \quad (21)$$

The single doublet determinant condition fixes

$$\xi_0^2 = 0.14616372381894022. \quad (22)$$

This decimal is an exported root of the matrix rank condition, not an input used to fit a measured mass. At the present high-scale tree-level gate the light eigenvalue is exactly zero; P2–P3 will replace that condition by the renormalized electroweak matching condition.

The full gradient residual is 3.80×10^{-16} . The radial solver finds 27 signed stationary roots; the desired $\omega = +1$ branch has $V = -0.87681703125$ and is the lowest enumerated branch. The Hessian has 38 zero modes and no negative modes at tolerance 8.50×10^{-7} . Direct projection accounts for 33 gauge modes, one PQ mode, and four light-doublet real modes. For the latter,

$$(C_3, C_2, Y^2) = (0, 3/4, 1/4), qquad \|h_{10}\|^2 = 0.9999920937, quad \|h_{126}\|^2 = 7.9063 \times 10^{-6}. \quad (23)$$

Thus it is one $(1, 2, \pm 1/2)$ real multiplet, not an accidental quartet.

The complete Standard-Model Casimir decomposition is below. “3-pair” and “6-pair” denote an irrep and its conjugate in the real Hessian. Equal masses on distinct rows are genuine degeneracies, not merged labels.

m^2	$SU(3)$	$\dim SU(2)$	$ Y $	real mult.
0	1	1	0	2
0	1	2	1/2	4
0	3-pair	2	1/6	12
0	3-pair	1	2/3	6
0	1	1	1	2
0	3-pair	2	5/6	12
0.0382803065	1	2	1/2	4

m^2	$SU(3)$	$\dim SU(2)$	$ Y $	real mult.
0.0888808155	1	1	0	1
0.1121882733	3-pair	2	1/6	12
0.1159441884	1	1	0	1
0.1866355208	3-pair	1	1/3	6
0.2116296538	3-pair	1	1/3	6
0.2600000000	8	1	0	8
0.2722422827	6-pair	1	2/3	12
0.3939174318	1	2	1/2	4
0.4006848071	3-pair	1	1/3	6
0.5400000000	1	3	0	3
0.5522408476	1	3	1	6
1.4210000000	6-pair	1	1/3	12
1.4210000000	3-pair	3	1/3	18
1.4210000000	3-pair	1	4/3	6
1.4700000000	6-pair	3	1/3	36
1.4700000000	6-pair	1	4/3	12
1.4700000000	1	1	2	2
1.5188026219	8	2	1/2	32
1.5188026219	3-pair	2	7/6	12
1.5210998141	1	2	1/2	4
1.5721973781	8	2	1/2	32
1.5721973781	3-pair	2	7/6	12
1.7576694975	3-pair	1	1/3	6
1.8266780732	3-pair	1	1/3	6
2.0580077173	6-pair	1	2/3	12
2.0580091524	1	3	1	6
2.1580617267	3-pair	2	1/6	12
2.8334249961	1	1	0	1

The real multiplicities sum to 328, and the maximum leakage of any mass eigenspace under the 12 unbroken generators is 1.93×10^{-10} . The independent vector result is

$$\frac{m_V^2}{g_{10}^2} : quad0^{\times 12}, \quad 0.1225^{\times 8}, \quad 0.6125^{\times 1}, \quad 1^{\times 12}, \quad 1.1225^{\times 12}. \quad (24)$$

The χ_7 -only Hessian has Frobenius norm 2.32379×10^{-2} , which explicitly verifies that the invariant-basis correction enters P1.

8 Acceptance and next gate

P1 has four logically separate outputs:

1. a complete stationary-branch ledger from the corrected action;
2. a 328×328 Hessian whose field count and symmetry are checked;
3. vector-by-vector Goldstone alignment for 33 gauge modes and one projected PQ mode;
4. the full scalar and vector eigenvalue multisets for the declared point.

The executed benchmark passes all four items: the verifier reports 14/14 checks, a semidefinite radial-global branch, and exactly the required light doublet. This closes the tree-level algebraic P1 gate. It does not assert loop stability, a measured-scale fit, or cosmological longevity outside the

enumerated singlet branches. The next mainline task is P2: run and match the actual spectrum, propagate parameter/scale covariance, and reimpose the light-doublet condition on the renormalized mass matrix.

References

- [1] K. S. Babu and S. Khan, “A Minimal Non-Supersymmetric SO(10) Model,” [arXiv:1507.06712](#).